

trades *physical* stocks, but Figure 7.4 involves trading *risk-neutral*, or option, volatility. Trading volatility in derivatives markets brings an additional volatility risk premium that rebalancing does not. Thus, losses in trading volatility in derivative markets are potentially much steeper than simple rebalancing strategies. Second, pure volatility trading in derivatives can be done without taking any stances on expected returns through delta-hedging. Rebalancing over fundamental asset or strategy positions is done to earn underlying factor risk premiums. While there is only weak predictability of returns, the investor practicing rebalancing gets a further boost from mean reversion as she buys assets with low prices that have high expected returns. Chapters 4 and 14 cover this in more detail.

Constructing valuation models with volatility risk can be tricky because the relation between volatility and expected returns is time varying and switches signs and is thus very hard to pin down. A large literature has tried to estimate the return–volatility trade-off as represented in equation (6.1) repeated here:

$$E(r_m) - r_f = \bar{\gamma} \sigma_m^2, \quad (7.1)$$

where $E(r_m) - r_f$ is the market risk premium and σ_m^2 is the variance of the market return. According to CAPM theory, $\bar{\gamma}$ represents the risk aversion of the average investor.

Is the coefficient, $\bar{\gamma}$, relating the market volatility or variance to expected returns, which is supposedly positive in theory, actually positive in data? In the literature, there are estimates that are positive, negative, or zero. In fact, one of the seminal studies, Glosten, Jagannathan, and Runkle (1993), contains all three estimates in the same paper! Theoretical work shows that the risk–return relation can indeed be negative and change over time.¹³ What is undisputed, though, is that when volatility increases dramatically, assets tend to produce losses. Only an investor who can tolerate large losses during high-volatility periods should consider selling volatility protection.

2.4. OTHER MACRO FACTORS

Several other macro factors have been investigated extensively in the literature and deserve attention from asset owners.

Productivity Risk

A class of *real business cycle models* developed in macroeconomics seeks to explain the movements of macro variables (like growth, investment, and savings) and asset prices across the business cycle. In these models, macro variables and asset prices vary across the business cycle as a rational response of firms and agents adjusting to *real shocks*. The label “real” in “real business cycle” emphasizes

¹³ See, for example, Backus and Gregory (1993), Whitelaw (2000), and Ang and Liu (2007).